DO NOW — RECALL

Lesson 143 · The TCP/IP Stack and Encapsulation

Name: _____ Class: _____ Date: _____

1. Which device typically forwards traffic between different IP networks?

2. Which logic gate outputs 1 only when both inputs are 1?

3. What is the accumulator for?

4. Which is faster in the worst case on sorted arrays, linear or binary search?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Router.

2. Model answer

AND.

3. Model answer

Holding a CPU's working data/ALU results.

4. Model answer

Binary search: O(log n) rather than O(n).

By the end, you can…

• HTTP sends a request body/application message.

• TCP segments carry ports, IP packets carry network addresses.

• Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Vocabulary — know these terms

application layer — Protocols providing services to programs, such as HTTP or DNS.

transport layer — End-to-end service such as TCP or UDP.

internet layer — IP packet addressing and routing between networks.

link layer — Local frame delivery on physical or wireless link.

NEW LEARNING

Read, annotate and explain

Application data

HTTP sends a request body/application message.

Transport and IP

TCP segments carry ports, IP packets carry network addresses.

Frame on local link

Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

HTTP application data

Identify the relevant information and make a prediction.

TCP segment in IP packet

Explain the change using correct technical terms.

Ethernet frame

HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name four layers of the TCP/IP stack and give one role for each. [8]

Question 2

Describe encapsulation of application data over TCP/IP and Ethernet. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain what a router changes when forwarding a packet to a different link technology. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every router preserves the same link-layer frame all the way to the destination.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.